Chapter 4: Web Services & APIs

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Syllabus

4.1 API basics: Role in web applications

4.2 REST principles and design, RESTful APIs

4.3 JSON versus XML data exchange

4.4 Data validation and serialization

4.5 Microservices

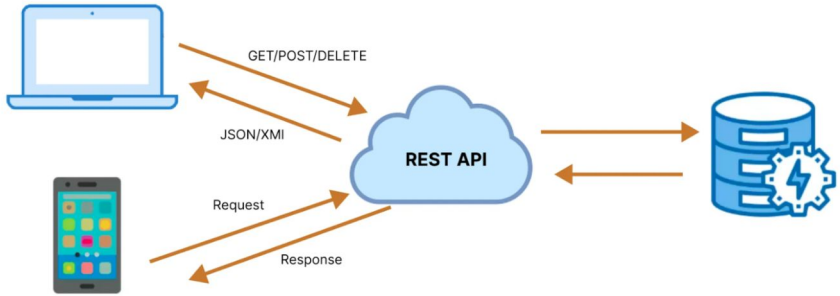
Total Marks: 9

Application Programming Interface (API)

- API stands for Application Programming Interface
- They enable applications and systems to communicate and share data efficiently.
  - Enable seamless interaction between applications, platforms, and services.
  - Power everyday features such as weather updates, payments, and e-commerce checkouts.
- Why we need API:
  - **Reusability**: Avoid reinventing the wheel by leveraging existing APIs (e.g., Google Maps API, Stripe Payments API).
  - **Efficiency**: Save development time by integrating ready made functionalities.
  - **Scalability**: Enable modular, distributed systems that can grow easily.
  - **Integration**: Connect multiple platforms web, mobile, IoT, or analytics.
  - **Automation**: APIs allow machines to talk to machines without manual input.

APIs Working

- **Request**: The client sends a request to an API endpoint (URI).
- **Processing**: The API forwards the request to the server.
- **Response**: The server processes and sends back the requested data.
- **Delivery**: The API returns the server's response to the client.



API Architectures

- **REST (Representational State Transfer)**:
  - REST is a simple, flexible API architecture that uses HTTP methods (GET, POST, PUT, DELETE) for communication.
  - Data Format: JSON, XML.
- **SOAP (Simple Object Access Protocol)**:
  - SOAP is a more rigid protocol that requires XML based messaging for communication. Strict and secure protocol using XML for structured messaging.
  - Data Format: XML
- **GraphQL:**
  - Modern query language that lets clients fetch only the data they need.
  - Data Format: JSON
- **gRPC:**
  - High performance framework using Protocol Buffers (Protobuf).
  - Data Format: Binary

Representational State Transfer (REST)

- REST stands for **Representational State Transfer**
- The constraints of REST architecture allowing interaction with RESTful web services.
- It defines a set of functions **(GET, PUT, POST, DELETE)** that clients use to access server data.
  - **GET (retrieve a record)**
  - **PUT (update a record)**
  - **POST (create a record)**
  - **DELETE (delete the record)**
- Its main feature is that REST API is **stateless**, i.e., the servers do not save clients' data between requests.

Core Principles of REST

- **Statelessness**:
  - The server does not "remember" the client's previous interactions.
- **Client-Server Separation:**
  - There is a clear separation of concerns.
  - The Client (browser/mobile app) handles the user interface and initiates requests, while the Server handles data storage, processing, and sends responses
- **Uniform Interface**
  - APIs should use a standard, consistent way of communicating.
  - This usually means relying on the standard **HTTP protocol** (GET, POST, etc.) rather than inventing new rules.
  - This makes the API scalable and easier to maintain.

RESTful API Design Guidelines

- **Resource Modeling (Naming Conventions)**
  - **Use Nouns, Not Verbs:** URIs should name the thing (resource), not the action. The action is defined by the HTTP method (GET, POST, etc.).
    - **Bad**: /getUsers or /updateUser
    - **Good**: /users
  - **Use Plural Nouns**: Use plural names for collections of resources.
    - Example: /users (Collection) vs. /users/:id (Single resource) .
  - **Use Lowercase**: Always use lowercase letters in URIs.
  - **Use Hyphens:** Use hyphens (-) to separate words, not underscores or CamelCase.
    - Example: /user-profile instead of /userProfile.
  - **Versioning**: Always version your API to manage changes without breaking client applications.
    - Example: /v1/users.
- **HTTP Methods** (The Verbs)
  - GET (retrieve a record)
  - PUT (update a record)
  - POST (create a record)
  - DELETE (delete the record)

RESTful API Design Guidelines

- **Standard Status Codes**
  - **Success**:
    - **200 OK**: Everything is fine.
    - **201 Created:** A new resource was successfully created.
  - **Client Errors (User's Fault):**
    - **400 Bad Request:** Invalid payload or input.
    - **401 Unauthorized:** Credentials are missing or incorrect .
    - **403 Forbidden:** You are logged in but don't have permission.
    - **404 Not Found:** The URL or resource does not exist.
    - **429 Too Many Requests:** Rate limiting exceeded.
  - Server Errors (API's Fault):
    - **500 Internal Server Error**: Syntax error or server crash.

Advanced Design Topics

- ## Filtering and Pagination
  - Use **Query Parameters** (the part of the URL after the ?) to **filter, sort, or paginate** data.
  - This allows clients to request only the specific data they need, improving performance.
  - **Example: ?page=1&sortBy=name&sortDir=asc**.
- ## Hypermedia
  - Include links to related resources within the API response to help discoverability.
  - Example: Returning a link to a user's photo **(https://.../images/1.jpeg)** inside the user object .
- ## Optimizing Performance
  - **Compression**: Use compression techniques to reduce data size.
  - **Nested Objects**: Return full resource objects instead of just IDs to prevent the client from making multiple extra calls.
    - **Bad**: Returning **"authorId": 1** (forces another call to get author name) .
    - **Good**: Returning **"author": { "id": 1, "name": "Author Name" } .**

JSON (JavaScript Object Notation)

- JSON is a lightweight **data-interchange format** based on a subset of the JavaScript programming language .
- It is designed to be **easy** for humans to **read and write**, while also being easy for machines to parse and generate.
- Syntax Structure
  - **Objects**: Data is enclosed in curly braces {} to define an object.
  - **Key-Value Pairs:** Data is stored as pairs, where a field name (key) is associated with a value.
    - Example: {"name": "username"}.

JSON

```json
{
  "postId": 1,
  "title": "some post title",
  "content": "post contend",
  "author": {
      "id": 2,
      "name": "author name"
  }
}
```

Hypermedia Example

JSON

```json
{
  "name": "username",
  "photo": {
      "fileName": "1.jpeg",
      "link": "https://storage.example.com/images/1.jpeg"
  }
}
```

XML (eXtensible Markup Language)

- XML stands for eXtensible Markup Language.
  - Definition: It is a **markup language** designed to **store and transport data.**
  - Role in API: Like JSON, it is a **standard format for API payloads (requests and responses).** It allows different systems to exchange information.
  - "Extensible": Unlike HTML (where tags like <h1> are fixed), in XML, developers define their own tags (e.g., <user>, <price>).
- XML Structure & Example
  - XML uses a tree structure consisting of nested tags.
    - **Tags**: Data is wrapped in start <tag> and end </tag> **markers.**
    - **Root Element:** Every XML document must have one single parent "root" element that wraps everything.
    - **Attributes**: Metadata added inside the opening tag (e.g., id="101").

```xml
<user>
  <id>101</id>
  <name>John Doe</name>
  <email>john@example.com</email>
  <roles>
    <role>Admin</role>
    <role>Editor</role>
  </roles>
  <isActive>true</isActive>
</user>
```

Parsing

"Parsing" is the process of converting the raw text received from the API into a usable object in the programming language.

Raw Data: **'{"title": "API Design", "price": 29.99}'** → It's in string

For Json

```
// The API sends a string
const jsonString = '{"title": "API Design", "price": 29.99}';

// We parse it into an object with ONE line
const book = JSON.parse(jsonString);

// We use the data immediately
console.log(book.title); // Output: API Design
```

Parsing

Raw Data: **'{"title": "API Design", "price": 29.99}'** → It's in string

For XML

```
// The API sends a string
const xmlString = '<book><title>API Design</title><price>29.99</price></book>';

// We need a special parser tool
const parser = new DOMParser();
const xmlDoc = parser.parseFromString(xmlString, "text/xml");

// We have to find the specific tag manually to get the data
const title = xmlDoc.getElementsByTagName("title")[0].childNodes[0].nodeValue;

console.log(title); // Output: API Design
```

Serialization & Deserialization

- Serialization (Server → Client):
  - Definition: This is the process of **converting complex data structures** (like a user object in a database or a class instance) **into a format that can be easily transmitted over the internet** (like a **JSON string**) .
  - Role in API: When the server sends an **API Response, it must "serialize" the data** found in the database into a JSON payload so the browser can read it .
  - Example: The document shows converting **raw data into a structured JSON response** to optimize performance .
- Deserialization (Client → Server):
  - Definition: This is the reverse process. It takes the **"raw text" (payload) sent by the client and converts it back into a complex object or data structure** that the server's code can understand and manipulate.
  - Role in API: When a **client sends a POST or PUT request with data**, the server "deserializes" that JSON body to save it to the database.

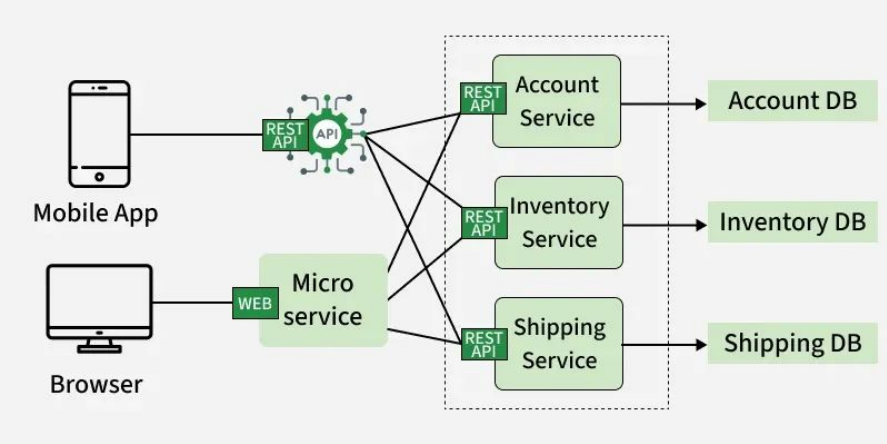
Data Validation

Before the server accepts deserialized data (e.g., to create a new user), it must **verify that the data is correct and safe.**

- What is it? The process of **checking the Payload (input data)** against a **set of rules.**
- Why is it needed? To **ensure** the data follows the **correct format before processing.** For example, **ensuring an email field** actually contains an email address.
- Handling Failure:
  - If validation fails, the API stops immediately and returns an error.
  - Status Codes: The document highlights 400 Bad Request for invalid payloads.
  - Specific Validation Errors: It also details a specific error response format (Status 422) for validation issues, providing details on what went wrong .

Microservices Architecture

- Microservices are an architectural approach to developing software applications **as a collection of small, independent services** that communicate with each other over a network.
- Components:
  - **Microservices**
    - Microservices are independent, loosely coupled services designed around specific business functions.
      - Handle a single, well-defined capability.
      - Can be developed and deployed independently.



Microservices Architecture

- **API Gateway**
  - The API Gateway serves as a centralized entry point for all external client requests.
    - Manages request routing and authentication
    - Forwards requests to appropriate microservices
- **Service Registry and Discovery**
  - Service Registry and Discovery keeps track of available services and their locations.
    - Stores service network addresses.
    - Enables dynamic inter-service communication.
- **Load Balancer**
  - A Load Balancer distributes incoming traffic across service instances.
    - Improves availability and reliability.
    - Prevents service overload.
- **Containerization**
  - Containerization packages microservices and their dependencies into containers.
    - Docker encapsulates services consistently
    - Kubernetes manages scaling and orchestration
- **Event Bus / Message Broker**
  - An Event Bus or Message Broker enables asynchronous communication between services.

Microservices Architecture

- **Database per Microservice**
  - In the Database per Microservice pattern, each microservice owns and manages its own dedicated database to maintain data autonomy.
- **Caching**
  - Caching improves performance by storing frequently accessed data closer to services.
    - Reduces database load
    - Decreases response latency
- **Fault Tolerance and Resilience**
  - Fault tolerance and resilience mechanisms enable the system to continue functioning even when some components fail.
    - Uses techniques such as circuit breakers, retries, and fallbacks.
    - Maintains overall system stability and availability.

Loose Coupling in Microservices

- **Definition**:
  - Loose coupling means system components (services) operate independently and do not rely heavily on one another's internal workings.
  - Changes to one service do not break the others or the client application.
- **Role of API Gateway:**
  - Achieved by using an API Gateway as an intermediary between the client and the backend services.
  - The Gateway acts as a "Gatekeeper" or proxy, ensuring the client never talks directly to the backend.
- **Client Independence:**
  - Clients only need to know the Gateway's endpoint, not the complex structure of the backend services.
  - Backend services can be updated, moved, or replaced without requiring changes to the client app.
- **Decoupled Logic:**
  - Common tasks like Authentication, Rate Limiting, and Usage Metrics are offloaded to the Gateway.
  - This keeps individual services focused purely on their specific business logic, rather than security or traffic management.

API Endpoints

- **Definition**: An endpoint is a specific URL that corresponds to a resource or set of resources.
  - Structure: It combines an HTTP method with a URI to define the action.
    - Example: GET /users to retrieve a list, POST /users to create a new user .
  - Best Practice: Use plural nouns for collections (e.g., /users) and avoid using verbs in the URL (e.g., avoid /getUsers) .
- **Returning JSON**
  - Body: The actual data requested (the payload).
  - Status Code: A numerical code indicating success (200 OK, 201 Created) or failure (400 Bad Request, 404 Not Found) .
  - Headers: Metadata about the response, such as Content-Type: application/json
- **Why API Testing is Critical**
  - The Challenge: API testing is considered one of the most challenging types of testing.
  - The Risk: Missing test cases can lead to significant problems in production after full integration.
  - Debugging: Issues found in production are much harder to debug than those found during initial testing.

API Testing Tools

Postman: Manual and automated API testing

SoapUI: SOAP & REST API testing

JMeter: Load and performance testing

Apigee: Enterprise API management

vREST: Automated regression testing